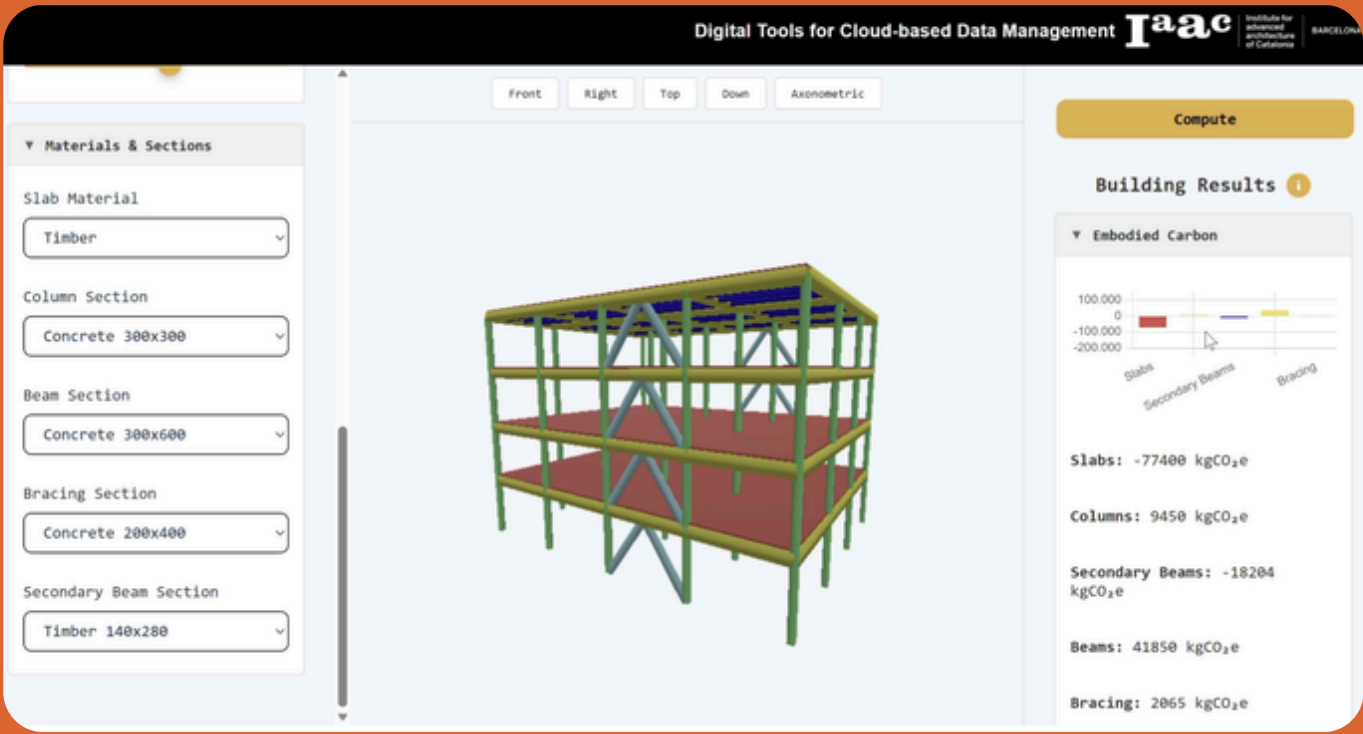


CARBON FRAME

EMBODIED CARBON WEB-APP



TYPE: Software Development

YEAR: 2026

TEAM: Andrea Cutroni

FACULTY: Justyna Szychowska,
Chuck Driesler, Sophie Moore

CONTEXT: IAAC Master in
Advanced Computational Design
(MACAD)

CarbonFrame is a web-based parametric tool for estimating the embodied carbon of structural frames. It allows users to define key geometric parameters, assign materials and sections from concrete, steel and timber, and generate real-time data on emissions, mass, and quantities. With comparison features and model export for further analysis, it supports early-stage decision-making and helps designers evaluate structural options with lower environmental impact.

PARAMETRIC STRUCTURAL FRAME

Number of Floors: 6

Floor Height (m): 4

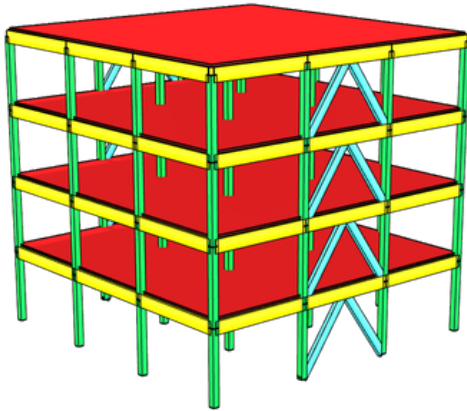
Bay Length (m): 5

Number of Bays: 3

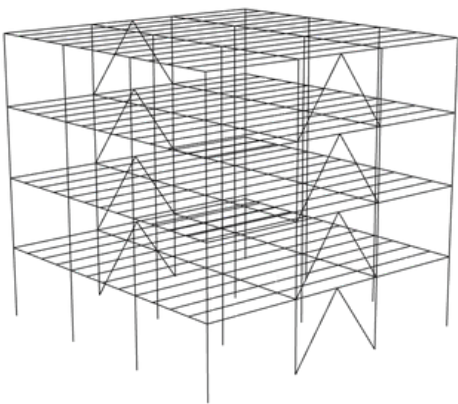
Slab Thickness (m): 0.1

Secondary Beam Spacing (m): 1

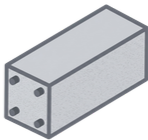
DETAILED MODEL



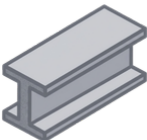
ANALYTICAL MODEL



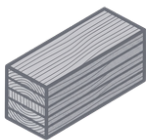
MATERIALS & SECTIONS



Concrete



Steel

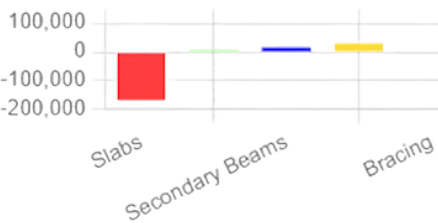


Timber

MASS CALCULATION



EMBODIED CARBON ANALYSIS

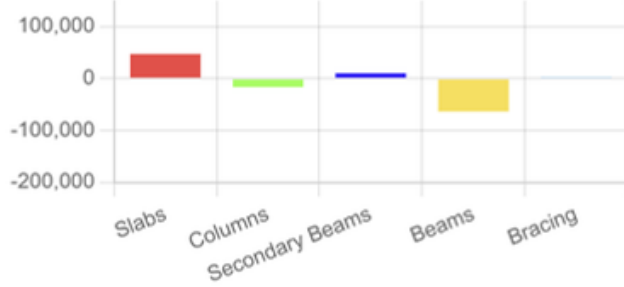


QUANTITY TAKEOFF

Columns: 64
Beams: 96
Secondary Beams: 108
Bracing: 16

COMPARISON MODE

CURRENT RUN



PREVIOUS RUN

